



SMART INDIA HACKATHON 2026

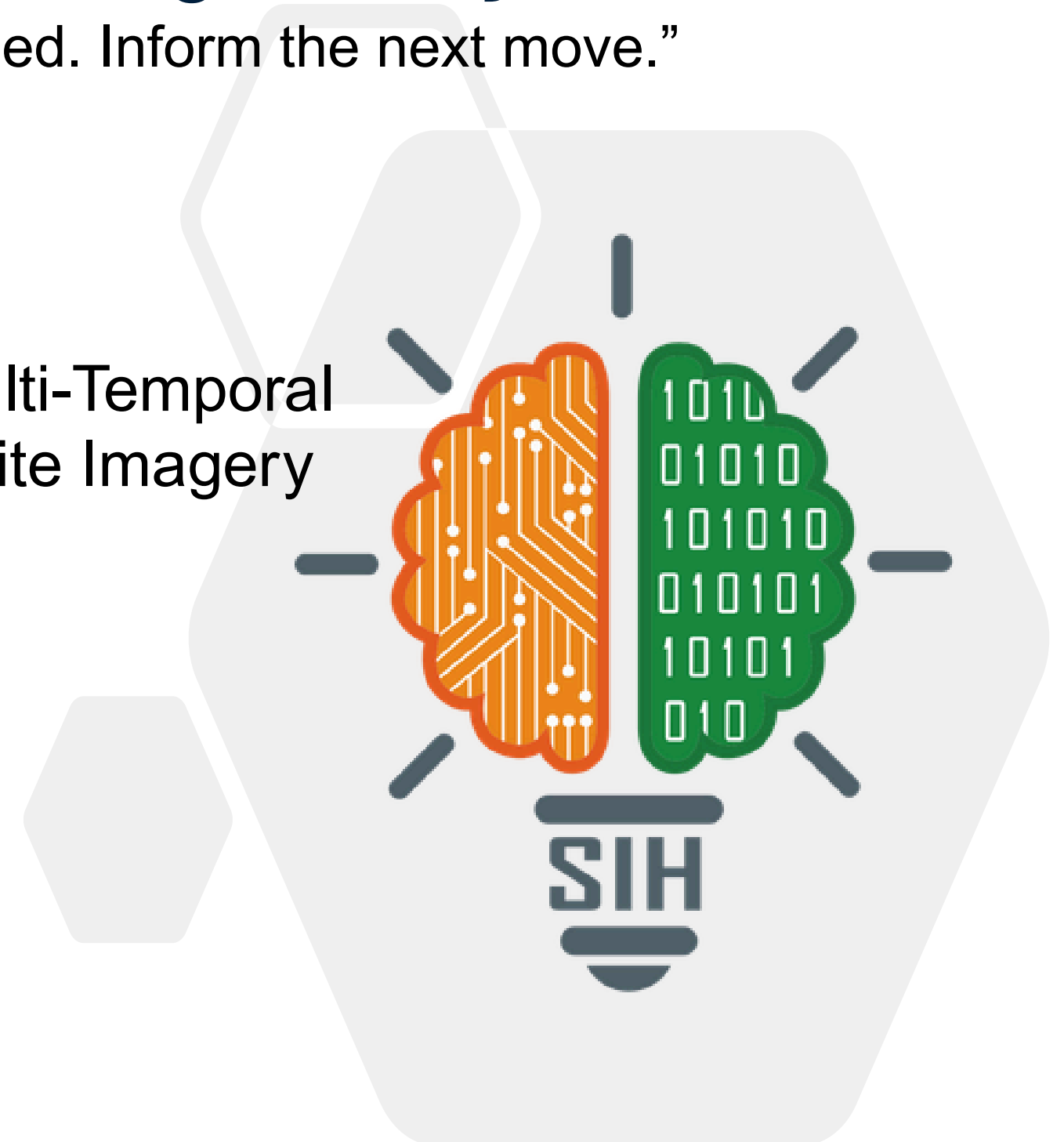
INNOVATION FOR A VIKSIT BHARAT



ASTREVA : Defence Satellite Intelligence System

“Describe what you seek. Trace what changed. Inform the next move.”

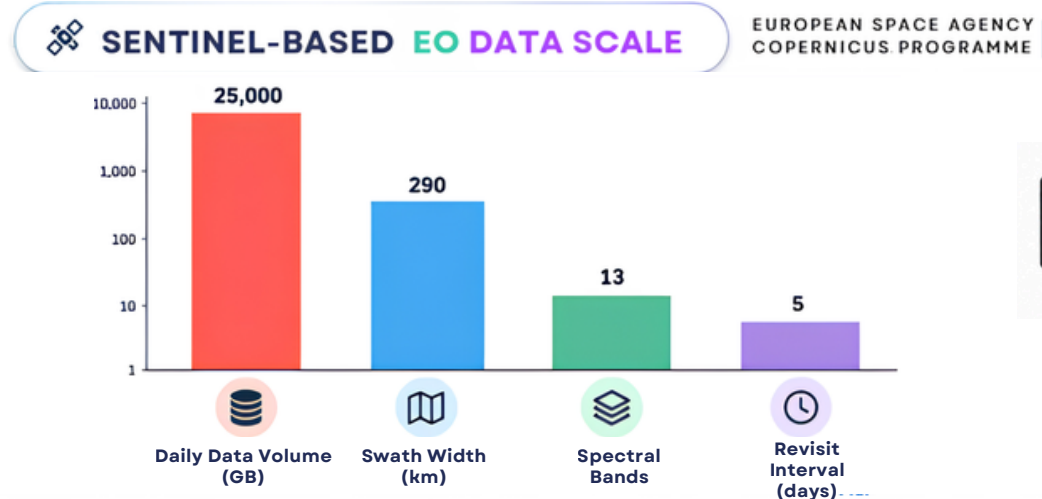
- **Problem Statement ID – SIH26227**
- **Problem Statement Title-** Semantic Retrieval and Multi-Temporal Change Analysis of Satellite Imagery
- **Theme-** Space and Technology
- **PS Category-** Software
- **Team ID-** 170575
- **Team Name -** Impacteers



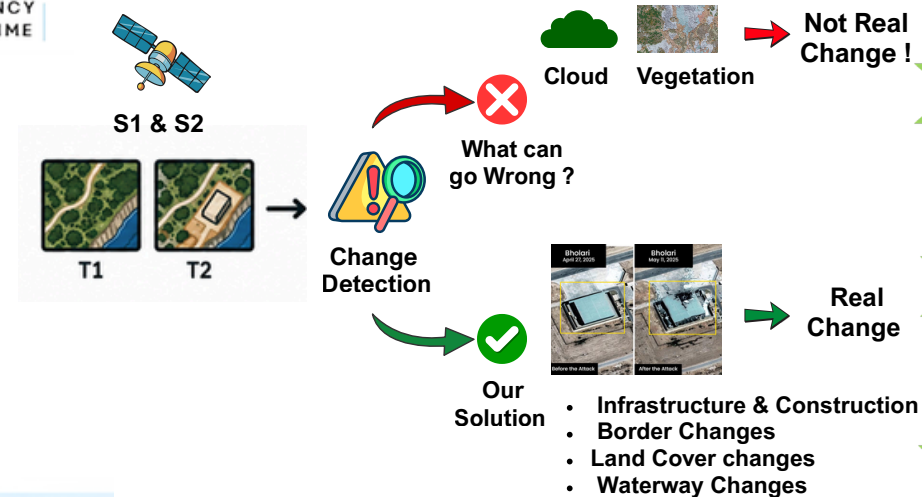
PROBLEM ADDRESSED



- Sentinel programme-scale data production can reach **25+TB/day**, making manual discovery across growing multi-temporal archives increasingly difficult.
- Sentinel-2's 5-day revisit cycle generates a rich temporal timeline, but weather, seasons and viewing conditions can make unchanged areas look different.



Source: ESA / Copernicus Sentinel Mission Documentation



PROPOSED SOLUTION



“We’re building the Intelligence Layer for India’s skies – one trusted system to discover, understand and verify what changed across space and time.”

- **01 – SEARCH BY MEANING, NOT JUST METADATA**
Ask in natural language → ASTREVA finds the relevant satellite scenes.
- **02 – DETECT REAL CHANGE, NOT JUST DIFFERENCE**
Before → After → Across Time → Verified Change.
Multi-temporal, multi-sensor analysis with quality masking, registration checks and false-change suppression
- **03 – ONE INTELLIGENCE LAYER, BUILT FOR DEFENCE**
From fragmented satellite archives → one searchable, traceable intelligence view. Designed for on-premises/offline operation

INNOVATION:

- **Watch Changes from Multiple factors**
Combines **optical, radar and terrain information** to detect changes that may be invisible to a single satellite image.
- **Know What Is Actually Changing**
Doesn't treat every difference as a change – it **filters out weather, seasons, shadows and image errors** to focus on genuine ground changes.
- **Search for Changes, Not Just Images**
What changed + Where + When → actionable intelligence

“The landscape changes. Satellite archives grow. **ASTREVA connects them – giving Defence a single intelligence view across space and time.”**



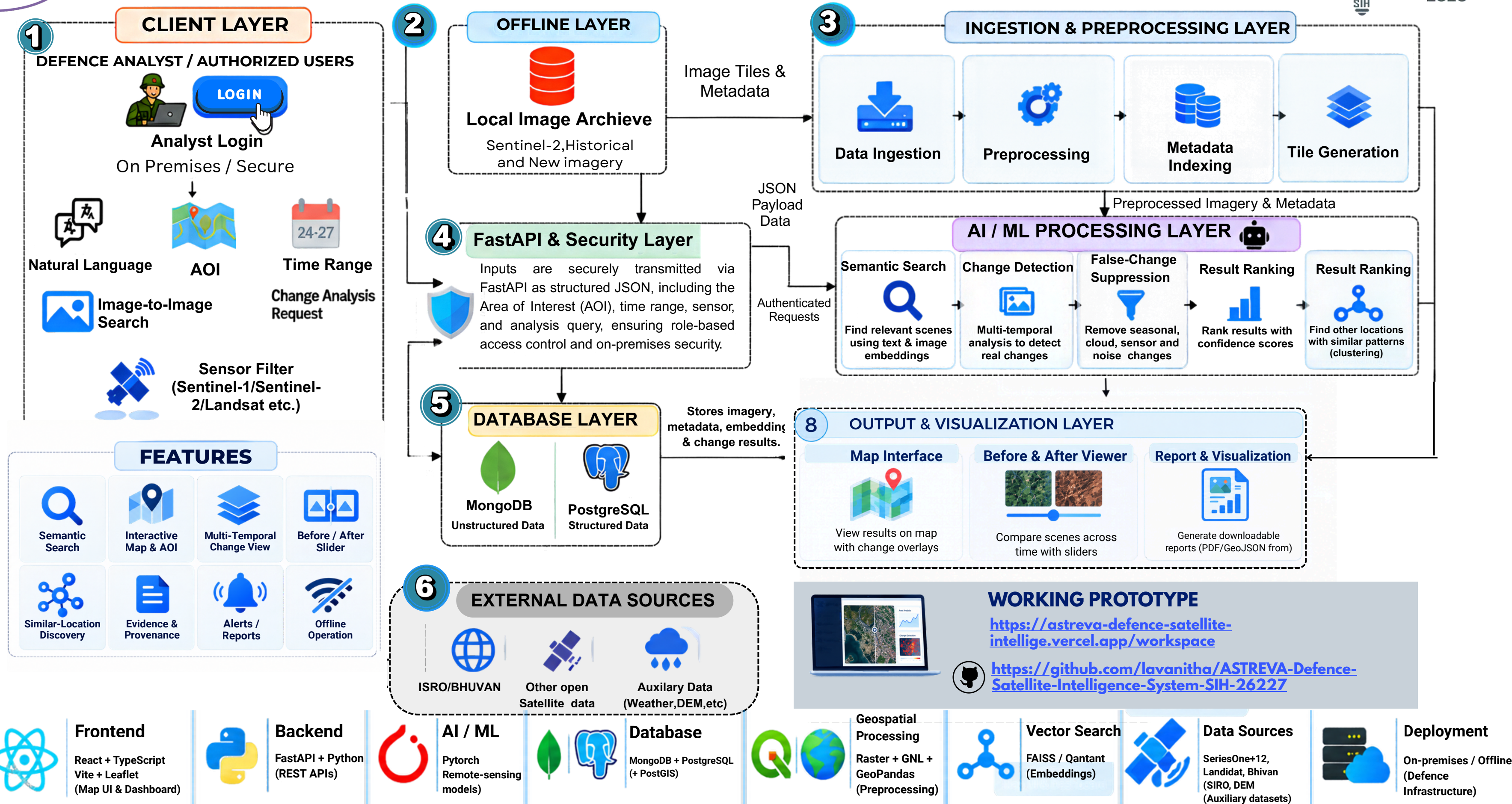
ASTREVA – PROTOTYPE OVERVIEW

https://youtu.be/7kuG_UILMkY?si=-4dtWzJL6DCiDnBK





TECHNICAL APPROACH



Feasibility



Technical → Offline-ready AI + geospatial processing → 100% accessible

Operational → Automated data pipeline + Incremental ingestion of new imagery

Financial → Open data + Open-source stack → near-zero cost

Viability



High Demand → Faster access to Large Satellite Image Archives

Defence Value Impact → Quicker search and change verification

Scalable → Strategic regions → wider coverage

Feasibility and Viability

Risks & Barriers



Archive Scale

Large archives increase storage and indexing load.



Offline Constraints

Models and data must work without external services.



Processing Load

Large imagery volumes require efficient indexing.



Data Pipeline Phases



Phase 1 - ACQUIRE
Collect satellite imagery + metadata



Phase 3 - INDEX
Creating searchable imagery and metadata indexes



Phase 4 - ANALYZE
Finding, comparing and verifying meaningful changes

Phase 2 - PREPARE
Preparing imagery for reliable change analysis



Impact Metrics



Faster discovery of relevant satellite imagery
↓ **70%+**
search effort reduced



Quick addition of new satellite imagery
↓ **50%+**
update time reduced



Accurate and meaningful change detection
↓ **60%+**
false alerts reduced



Traceable and reliable imagery and results
↑ **100%**
source and history maintained

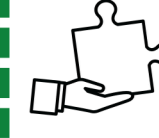
Strategies for Overcoming Challenges



Archive Scale: “Update, Don’t Rebuild”
Incremental indexing adds new imagery without rebuilding the archive.



Offline Constraints: “Stage Once → Run Offline” Models, data and indexes remain locally available without external services.



Processing Load: “Process Smart, Not Everything” Tiled processing and efficient indexing reduce unnecessary computation.

Key Stakeholders



Indian Army / DGIS
→ military personnel /
field intelligence icon



Defence Intelligence Analyst
→ analyst at multi-monitor
satellite-imagery workstation

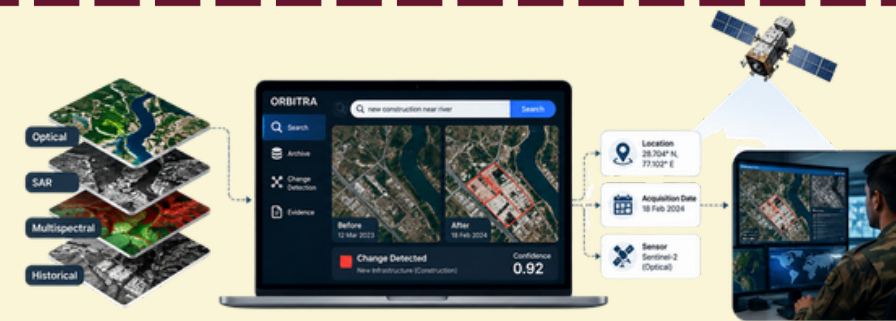


MoD & Strategic Agencies →
government institution / Ashoka
emblem



ISRO / NRSC & Data Providers
→ Earth-observation satellite
over India

Projected Defense Impact



Defence-Ready Operations - Secure, on-premises and offline-ready for controlled environments.

Projected impact based on
system capabilities,
SIH26227 requirements &
pilot workflow assumptions
(2026)


70%
**Faster
Intelligence Discovery**

Relevant satellite evidence
found faster through
semantic and multimodal
search.


60%
**Faster
Change Verification**

Multi-temporal comparison
reduces manual before/after
analysis and helps identify
meaningful changes quickly.


50%
**Reduced
Processing Effort**

Incremental indexing avoids
unnecessary archive-wide
reprocessing, saving
compute time and resources.


100%
**Complete Evidence
Traceability**

Every result retains its
source, acquisition time,
sensor and processing
history for auditable and
defensible intelligence.

Key Projected Impacts and benefits

01 Faster intelligence discovery
across the pilot region

02 More frequent optical
change monitoring

03 Recurring SAR-based
change verification

04 One integrated satellite-
intelligence workflow

Benefits

01 520.8 km² of the pilot AOI can
be indexed for continuous
satellite intelligence analysis

02 5-day Sentinel-2 revisit enables
frequent optical change
monitoring

03 6-day Sentinel-1 revisit adds
recurring SAR-based evidence,
including all-weather radar
observations

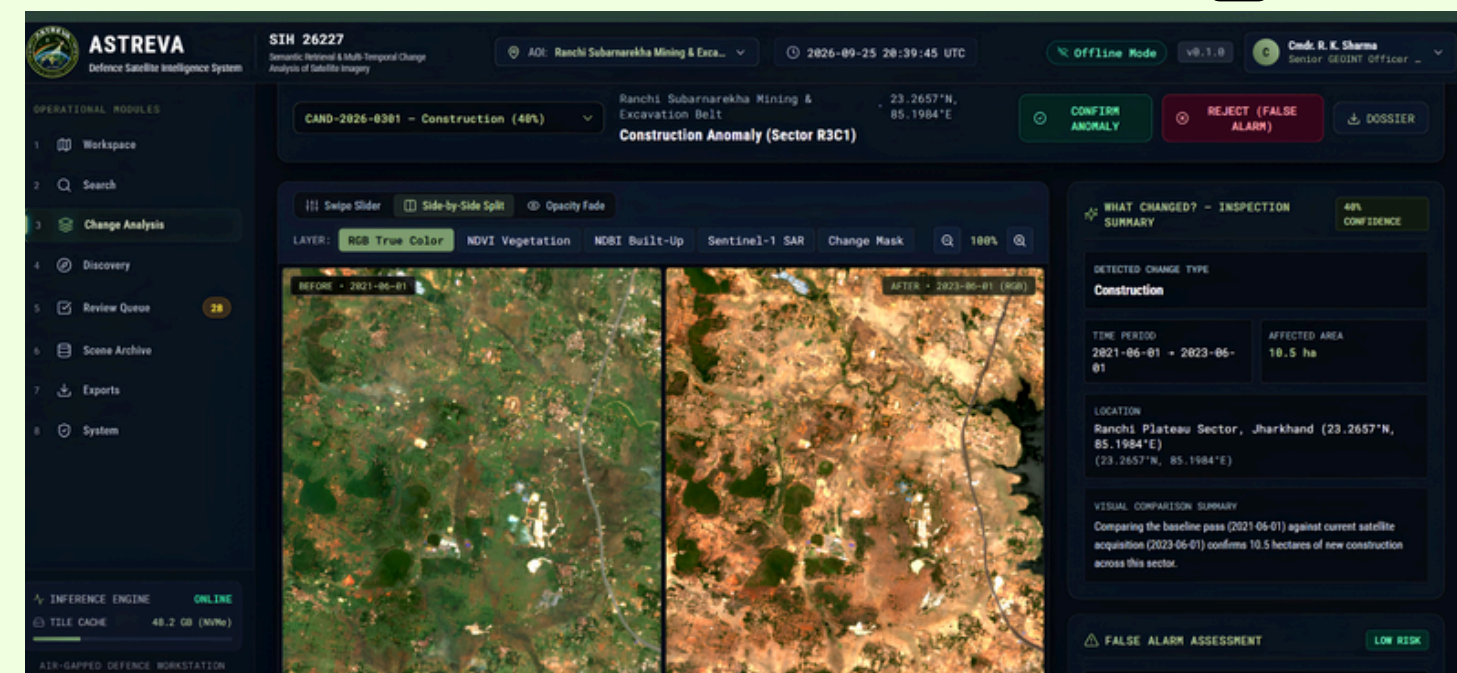
04 6 core intelligence capabilities
unified into one searchable,
change-aware workflow

**Key
Impacts**

Prototype Preview


1 Workspace – Area of Interest

2 Semantic Search & Retrieval

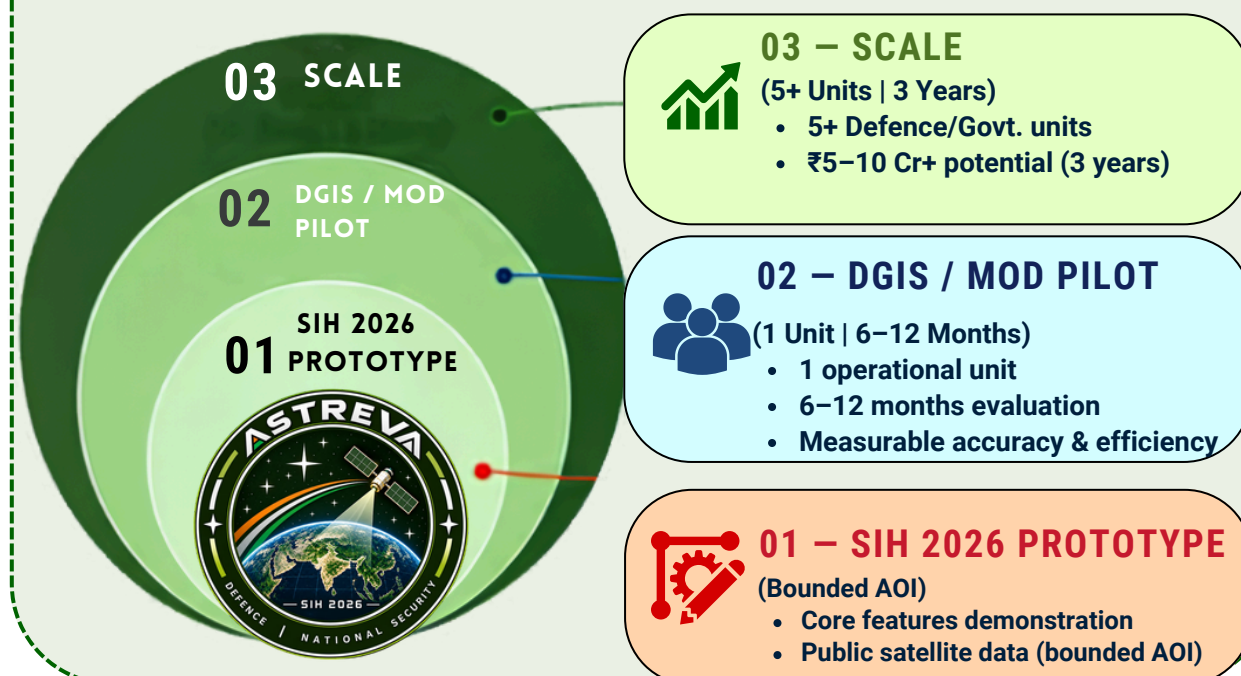
3 Change Analysis




WHY ASTREVA / RESEARCH

- 1** Conventional EO catalogues search by coordinates, date, platform and product metadata, analysts may still need to know where and when to look.
- 2** ASTREVA adds semantic text + image-to-image retrieval, multi-temporal change analysis, discovery/clubbing and analyst review with provenance.
- 3** The SIH brief requires on-premises operation with network access disabled during evaluation, incremental ingestion, georegarding/provenance and suppression of seasonal, atmospheric, geometric and sensor-induced false change.

ASTREVA GO-TO-MARKET (GTM) STRATEGY



ASTREVA – PROTOTYPE OVERVIEW

https://youtu.be/7kuG_UILMkY?si=-4dtWzJL6DCiDnBK

CAPABILITY COMPARISON

Capability	Google Earth Engine (Google cloud infrastructure)	Planet Insights (Cloud-native Cloud-heated)	Sentinel Hub (Cloud-based API service)	ASTREVA (Offline / On-premises evaluation)
Satellite Imagery analysis	✓	✓	✓	✓
Multi-temporal analytics	✓	✓	✓	✓
Semantic/ image retrieval	✗ Not core	✗ Not core	✗ Not core	✓
Air gaps on-premises runtime	✗ Cloud only	✗ Cloud only	✗ Cloud only	✓ Designed for offline on-premises
Incremental local vector index	✗ Not core	✗ Not core	✗ Not core	✓ Supported
Analyst provenance/audit	✗ Limited	✗ Limited	✗ Limited	✓ Signed audit / provenance

OPERATIONAL USE



ASTREVA – SDG ALIGNMENT & IMPACT

Environmental
Change MonitoringUrban Change
DetectionLand & Vegetation
MonitoringGeospatial Intelligence
& Innovation

REFERENCE LINKS



RemoteCLIP – Vision-Language
Foundation Model for Remote Sensing
<https://arxiv.org/abs/2306.11029>



STAC – OGC SpatioTemporal Asset
Catalog Standard
<https://www.ogc.org/standards/stac/>



Prithvi-EO-2.0 – A Versatile Multi-Temporal
Foundation Model for Earth Observation
Applications
<https://arxiv.org/abs/2412.02732>